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The role of transient rheology in polymeric sintering

Received: 30 March 2005
Accepted: 25 August 2005
Published online: 25 September 2005
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Abstract The initial theory of Frenkel and Eshelby for the coalescence of drops in air (or sintering) of Newtonian fluids, which equated the work of surface tension to the work done by viscous stresses while assuming biaxial extensional flow kinematics, was extended to the case of time-dependent material functions using the Upper Convected Maxwell (UCM) model. A numerical scheme was developed to solve the ordinary differential equations (ODE) for the stresses, which are embedded in the ODE based on the mechanical energy balance. Initial conditions required to solve the set of non-linear ODEs were obtained from visualization experiments of

the coalescing drops as the theory for elastic contact gave unrealistically high values of the initial neck radius. The transient model predicted that coalescence was accelerated by increasing the relaxation time, the opposite relationship of what was predicted by the steady-state UCM formulation, and was capable of quantitatively predicting the experimental coalescence rates at times when viscoelasticity was important.

Keywords Sintering · Coalescence · Polypropylene · Transient viscosity · Upper Convected Maxwell model

Introduction

Viscous coalescence refers to the process where, in an attempt to minimize surface area, surface tension drives a collection of fluid drops in air to merge into a single, homogeneous body. This process is also referred to as sintering, which is carried out in air rather than in the presence of another viscous fluid. Coalescence begins when drops are brought into contact, and they instantaneously deform in an elastic manner to create finite contact surfaces (Johnson et al. 1971). It is important to emphasize that elastic contact scales with material properties and particle radius but not time (Mazur et al. 1995). Elastic contact may also be referred to as neck formation, because it defines the formation of a bridge at the interface between particles. Coalescence continues

with the radial growth of the neck region. At this stage, the particles retain their individuality because neck growth is not influenced by other contacts with neighboring drops. In addition, there is no appreciable change in density of the multi-particle structure (Mazur et al. 1995). The neck dimensions grow until a network of pores forms and eventually become entrapped bubbles. The process of eliminating the entrapped bubbles is referred to as densification, and the densification rate depends upon gas permeability in the molten material (Mazur et al. 1995). A complete description of the coalescence of multiple drops consists of at least three stages: elastic contact, neck growth, and densification. However, the neck growth process is the focus of this work because it embodies most of the geometrical change experienced by a fluid particle of relatively low

viscosity (i.e., $< 10^6$ Pa s) during coalescence. Furthermore, it can be emphasized that the interest of this study is in coalescence of drops in the presence of air rather than another viscous fluid, such as, the studied by Verdier and Brizard (2002).

Frenkel (1945) was the first to propose an expression that described viscous (Newtonian) coalescence, which was later corrected for continuity by Eshelby (1949). They suggested that coalescence could be modeled by evaluating two isolated particles, which they represented by the idealized system of spherical drops as depicted in Fig. 1. In the figure, a_o is the initial particle radius, x is the neck radius, y is the distance from the particle center to the contact plane, and θ is the angle between the axial plane of symmetry and the line from the particle center to the neck radius, otherwise referred to as the coalescence angle. They derived their model by performing a mechanical energy balance, which resulted in the work of surface tension equaling the work done by the rheological stresses. The Frenkel model defines a relationship between material properties, particle size, and the rate of viscous neck growth. This relationship has been used as a qualitative benchmark for experimental data, demonstrating good agreement between measured values and the prediction that the growth of the dimensionless neck radius (x/a) is proportional to the square root of time (Bellehumeur et al. 1998). It should be noted that this relationship is based on small angles or short initial times and may not be valid for long times as the dimensionless neck radius approaches 1.0.

Although reasonable agreement between predictions of the Frenkel/Eshelby model and experimental coalescence data for Newtonian fluids was observed, there are several approximations in the theory, which need to be

considered before applying it to polymeric fluids. First, the particle radius was assumed to be constant. Pokluda et al. (1997) improved the accuracy of the Newtonian model by deriving an expression, which accounted for the change in particle radius. Second, the viscosity was assumed to be Newtonian. It was shown by Bellehumeur et al. (1998) that the Newtonian model overpredicted the coalescence rate and, hence, did not work well for polymeric materials. They attributed the model's deviation from their data to viscoelasticity and developed a viscoelastic coalescence model using the Upper Convected Maxwell (UCM) constitutive model in Frenkel's approach along with Pokluda's expression for particle radius. As a first approximation, they assumed that the stresses were at steady state at any instant during the coalescence process. Using increasingly large values for the relaxation time that were not based on experimental data, they found that the steady-state UCM coalescence model was in close agreement with their experimental data for materials with low Deborah numbers (De). Finally, in the Frenkel/Eshelby approach and its extension to the steady-state UCM model it was assumed that the flow was homogeneous biaxial extensional. Results from finite element solutions for the coalescence of fluids described by Newtonian and UCM constitutive relations (with low Deborah numbers) suggest that this assumption may be reasonable because the predicted streamlines indicate a biaxial extensional flow field in the bulk of the drop with deviations occurring primarily in the neck region (Hooper et al. 2000; Jagota et al. 1988; Martinez-Herrera and Derby 1995; Pokluda et al. 1997).

Before adding the complexities of a full numerical solution of the equations of change, it seems worthwhile in addressing the role of viscoelasticity, that the extension of the Frenkel/Eshelby approach be considered first. Furthermore, because full numerical solutions, such as those based on the finite element method, fail to converge when viscoelasticity becomes important, then obtaining meaningful results beyond those of the Newtonian case will be difficult. Hence, it could be difficult to identify the effects of viscoelasticity at first using a full numerical solution (Doeringhaus and Baird 2003).

Unfortunately, there is evidence that suggests that the steady-state UCM model does not accurately describe the coalescence behavior of all viscoelastic materials. The steady-state UCM model predicts that coalescence rates decrease with an increase in the relaxation time. This predicted behavior was used to suggest that viscoelasticity was responsible for the observation that two propylene-ethylene copolymers coalesced at a slower rate than predicted by the Newtonian model (Bellehumeur et al. 1998). However, it was shown that coalescence times for polytetrafluoroethylene (PTFE) and several acrylic resins were significantly shorter than what was predicted by the Newtonian model (Mazur et al. 1995; Mazur and Plazek 1994). It was stated that viscoelasticity was responsible for

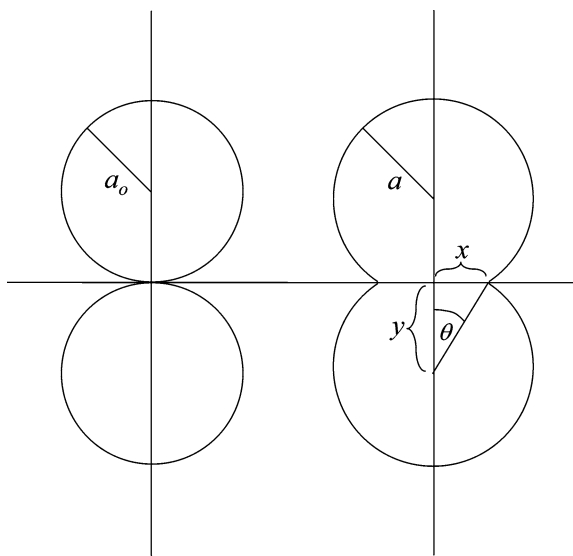


Fig. 1 Shape evolution during the coalescence of two spherical particles

this behavior as well, yet it is the opposite of what is predicted by the steady-state UCM model. Although the source for this discrepancy has not been identified, it is suspected that the steady-state approximation may not be valid for materials with large Deborah numbers (or at least in cases where the coalescence time and the longest relaxation time in the distribution are similar) and a complete transient representation is required. However, it should be noted that although it was necessary to arbitrarily pick the relaxation time to quantitatively predict the coalescence time, the time dependence of the dimensionless neck growth was consistent with viscous sintering and not with other modes (e.g., ‘zipping,’ which is an elastic driven response, which scales with $t^{1/7}$).

The objective of this work is to extend the approach proposed by Frenkel to describe the coalescence of two particles to the transient viscoelastic case by using the UCM constitutive equation without the assumption of steady-state stress behavior. The ability of the model to predict coalescence is assessed by comparing the measured neck growth values of three polymers with different mean relaxation times to the predicted values. The significance of this work rests partially on the fact that all of the material parameters in the UCM model are obtained directly from rheological data rather than arbitrarily adjusted to obtain agreement with the coalescence data. Furthermore, the coalescence of spherically shaped particles, on which the theory is based, is observed optically. Finally, the results presented here are used to explain the predictions of the steady-state model concerning the role of viscoelasticity in polymer coalescence.

Experimental

Materials

Isotactic polypropylenes (available from Sigma-Aldrich [CAS 9003-07-0]) of three different molecular weights were selected for this study. Because the materials possess different molecular weights, they are expected to have different relaxation times, potentially allowing the evaluation of the effect that the magnitude of the relaxation time has on coalescence. The average molecular weight, M_w , the polydispersity index, and the melt flow index are summarized in Table 1. The densities and the melt transitions for all samples were the same and were reported by the manufacturer as 0.9 g/cm³ and 160–165°C, respectively. The samples are referred to by their M_w values (190, 250, and 340 k).

Surface tension measurement

The surface tension of each of the materials was determined by fitting the Bashforth and Adams equation to

Table 1 Weight average molecular weight, polydispersity, and melt index of the three polypropylenes

$M (\times 10^{-3})$	PDI (M/N)	MI (g/10 min)
190	3.80	35.00
250	3.73	12.00
340	3.50	4.00

the sessile drop profile of the molten polymer in an inert atmosphere at 180°C (Padday et al. 1969). Sessile drops were prepared by first extruding polypropylene fibers with a Göttfert Rheograph 2001 capillary rheometer equipped with a 0.5-mm capillary die ($L/D = 10$). Pellets, with a diameter of approximately 500-μm and an aspect ratio of approximately 1, were cut from the fibers. A single pellet was placed vertically, standing on its cut end, on a glass slide in a Linkham hot stage set at 180°C, where it was melted into a sessile drop. The sample was quenched and the glass slide was rotated to allow a profile view of the drop. The sample was reheated to 180°C and a digital image of the profile was recorded with an optical microscope equipped with a miniDV camcorder. The accuracy of this technique (0.1% error) demands that the particle radii must be small so that gravitational forces cannot influence the shape of the profile. This was verified by calculating the bond numbers $Bo = \frac{\rho r^2 g}{\Gamma}$ ($Bo_{190k} = 0.063$, $Bo_{250k} = 0.125$, $Bo_{340k} = 0.063$), and supported by the observation that the profile shape did not change when the glass slide was rotated.

Rheological characterization

Rheological characterization was performed with a Rheometrics Mechanical Spectrometer Model 800 (RMS-800) and a Rheometrics Stress Rheometer Model 8600 (RSR-8600). For both instruments the test geometry was a 25-mm diameter cone and plate geometry with a 0.1-rad cone angle. All experiments were performed in the presence of an inert nitrogen atmosphere to prevent thermo-oxidative degradation. Test specimens were prepared by compression molding at 180°C under nominal pressure and allowing them to slowly cool without applied pressure. This method produces homogeneous samples with minimal residual stress. Reported rheological results represent the average of at least three runs using different samples for each run.

Using the RMS-800 stress growth upon inception of steady shear-flow experiments were conducted to obtain single-mode UCM model parameters. These experiments were performed at shear rates in the zero shear viscosity limit at 180°C. Small amplitude dynamic oscillatory shear measurements were performed to determine parameters

for the multimode UCM model. The tests encompassed frequencies from 0.1 to 100 rad/s at 180, 220, and 260°C. The results from the experiments performed at 220 and 260°C were then shifted to the coalescence temperature, 180°C, by applying time–temperature superposition to include frequencies lower than that can be obtained by direct measurement. The calculated errors for the stress growth and dynamic oscillatory measurements were found to be $\pm 10\%$ and $\pm 5\%$, respectively.

Creep measurements were carried out on the RSR-8600 to obtain the equilibrium creep compliance, J_e^o . These measurements were performed at shear stress levels, which were in the zero shear viscosity limit at 180°C. A constant shear stress was applied to all materials until steady-state creep was reached, $J(t) = J(\infty)$, after which the shear stress was removed and the reverse deformation (i.e., recoverable strain) was measured after the polymer was allowed sufficient time to relax. The ultimate creep recovery was used as direct measure of J_e^o (Macosko 1994). The repeatability of these measurements based on three repeat runs was found to be $\pm 6\%$. The values of J_e^o are listed in Table 2 for each resin.

UCM model parameter fitting

The single-mode UCM model parameters were obtained from stress growth data by minimizing the sum of the squared difference between the predicted and experimental transient viscosity. To represent the transient response, fitting was performed at 0.5 s intervals from inception of flow until steady state was achieved. Multimode UCM model parameters were obtained by simultaneously fitting Eqs. (1) and (2), using a nonlinear regression method, to storage (G') and loss (G'') modulus data as outlined in Bird et al. (1987):

$$G'(\omega_j) = \sum_{k=1}^N \frac{\eta_k \lambda_k \omega_j^2}{1 + (\omega_j \lambda_k)^2}, \quad (1)$$

$$G''(\omega_j) = \sum_{k=1}^N \frac{\eta_k \omega_j}{1 + (\omega_j \lambda_k)^2}, \quad (2)$$

where ω_j is the frequency of the j th data point and N is the number of modes. The optimum number of modes for each sample was determined by adding modes until

there was less than 10% reduction in the total error between the data and model.

Coalescence experiments

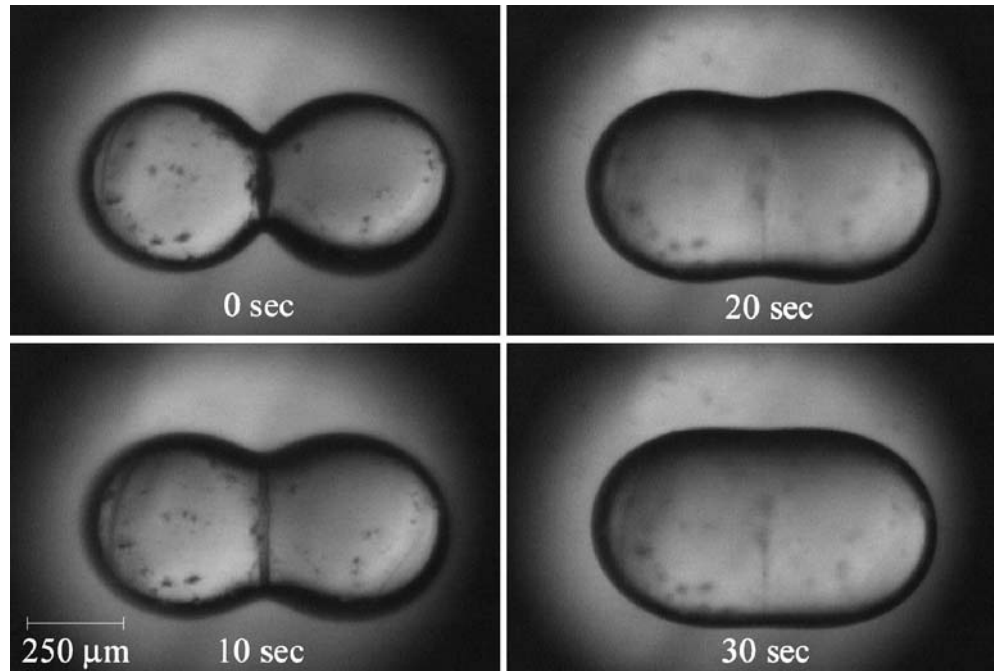
The sample preparation for the coalescence experiments was identical to the procedure used to generate sessile drops. Two of the pellets were placed vertically, standing on their cut end, next to each other in the hot stage at 180°C. The hot stage was capable of achieving a heating rate of 90°C per minute and could maintain the temperature within 0.1°C, which assisted in providing nearly isothermal conditions. An inert nitrogen atmosphere was used to help eliminate thermo-oxidative degradation throughout the experiment. Upon heating, the cylinders adopted a spherical shape before coalescence began. When observed from above with a microscope, the system geometry was identical to that shown in Fig. 1. The entire coalescence process was recorded on high resolution video. Still images were extracted at prescribed intervals, and the neck and particle radii were measured using Scion Image, a digital image analysis software available from Scion Corporation. Coalescence experiments were performed both with and without a lubricant (Dow Corning 200 R Fluid, 20 cst), in order to explore the importance of surface contact with the glass slide, but no difference was observed. As the possibility that surface contact with the glass slide could influence the neck growth results, the silicone lubricant was used to reduce the degree of wetting of the glass slide by the polymer. Each coalescence experiment was conducted three times to ensure reproducibility, and the reported neck radius versus time data is the average of the three runs. The standard deviation in the dimensionless neck radius between the repeated runs was 0.01.

A representative example of the recorded images is shown in Fig. 2, where there is initially a finite contact area and the neck radius increases with time until the two drops converge. The finite contact area is the result of a combination of elastic contact (which is minimal as discussed later) and some viscous growth as the sample is heated to 180°C. It was virtually impossible to obtain true initial values of the neck radius for these fluids primarily as there was some growth during the time the sample was heated from its melting point to 180°C.

Table 2 Single-mode UCM coalescence model parameters and calculated values for the Deborah number at 180°C. All values were also used for the Newtonian with the exception of the relaxation time

	λ (s)	η (Pa s)	J_e^o (1/Pa)	$\Gamma \pm 0.002$ (J/m ²)	$a_o \pm 1$ (μm)	De ($\lambda\Gamma/\eta_o a_o$)
190 k	0.18	1097.2	8.0E-4	0.024	232	0.017
250 k	0.73	3986.8	1.033E-3	0.015	232	0.012
340 k	1.54	11128.9	1.182E-3	0.028	274	0.014

Fig. 2 Optical micrographs from the coalescence of 190 k polypropylene drops



In the example, the two particles nearly reach a dimensionless neck radius of 1.0 within 30 s. Although the test is stopped a few seconds later, because the magnitude of the change in the dimensionless neck radius becomes comparable to the magnitude of the error in the measurement, the two particles do appear to continue to coalesce towards a single, nearly spherical drop.

Coalescence models

Newtonian model

The approach used by Frenkel to develop the viscous coalescence model began with performing a mechanical energy balance that equated the work of surface tension to the work done by the rheological stresses, which is referred to as viscous dissipation for a purely viscous fluid (Bird et al. 1987),

$$-\Gamma \frac{dS}{dt} = \tau : \mathbf{D} dV, \quad (3)$$

where τ is the deviatoric stress tensor, $\mathbf{D}$ is the rate of strain tensor,

$$\mathbf{D} = \frac{1}{2} (\nabla \mathbf{v} + \nabla \mathbf{v}^T), \quad (4)$$

where S is the surface area, and V is the volume of both spheres. The presence of external stresses and gravitational effects were neglected. The flow kinematics were

assumed to be biaxial stretching flow, with the components of the velocity gradient tensor given by,

$$\nabla \mathbf{v} = \begin{bmatrix} \dot{\epsilon} & 0 & 0 \\ 0 & -2\dot{\epsilon} & 0 \\ 0 & 0 & \dot{\epsilon} \end{bmatrix}, \quad (5)$$

where $\dot{\epsilon}$ is the extension rate. It can be noted here that Verdier and Brizard (2002) using visualization methods observed that in the interior of two drops coalescing in the presence of a viscous fluid the flow was primarily biaxial extensional flow. The rheological stresses in (3) were assumed to be described in terms of the Newtonian constitutive equation. At this point, Pokluda et al. (1997) applied the conservation of mass with constant density to account for the time dependence of the particle radius,

$$a(t) = a_0 \left(\frac{4}{[1 + \cos \theta(t)]^2 [2 - \cos \theta(t)]} \right)^{1/3}, \quad (6)$$

where θ is referred to as the coalescence angle, $x/a = \sin \theta$, as shown in Fig. 1. This time-dependent radius was incorporated into the mechanical energy balance, (3), through the definition for the surface area of the two spheres:

$$S = 4\pi a^2 (1 + \cos \theta). \quad (7)$$

As outlined in Pokluda et al. (1997), this system of equations (i.e., 3–7) can be rewritten as the following first-order differential equation:

$$\frac{d\theta}{dt} = \frac{\Gamma}{\eta a_o K_1^2} \frac{2^{-5/3} \cos \theta \sin \theta}{[2 - \cos \theta]^{5/3} [1 + \cos \theta]^{4/3}}, \quad (8)$$

where Γ is the surface tension, η the viscosity, and K_1 arises from the definition of the extension rate,

$$\dot{\epsilon} = -\frac{1}{2} \frac{\partial v_y}{\partial y} = K_1 \frac{d\theta}{dt}, \quad (9)$$

and is defined by:

$$K_1 = \frac{\tan \theta}{2} - \frac{\sin \theta}{6} \left[\frac{2(2 - \cos \theta) + (1 + \cos \theta)}{(1 + \cos \theta)(2 - \cos \theta)} \right]. \quad (10)$$

It should be added that the K_1 presented here is a slightly different form than was defined in the original derivation, which was an approximation in the limit of small coalescence angles. The K_1 defined here is valid over all coalescence angles during the coalescence process and should be somewhat more accurate over most of the neck growth process.

Steady-state UCM model

Bellehumeur et al. (1998) evaluated the Newtonian coalescence model, (8), but with K_1 based on small angles, for several polyethylenes and found that the predicted coalescence rates were faster than their experimental values. They proposed to incorporate viscoelasticity in the Newtonian coalescence model by using the steady-state version of the UCM constitutive equation for the extra stress tensor. By assuming that the flow field is homogeneous, and that at any instant the stresses are at steady state (though the stresses may change with the extension rate throughout the process) the constitutive equation was simplified to the following expression (Bellehumeur et al. 1998):

$$\tau + \lambda [-\nabla \mathbf{v}^T \cdot \tau - \tau \cdot \nabla \mathbf{v}] = 2\eta \mathbf{D}, \quad (11)$$

where λ is the characteristic relaxation time. After (5), (6), (7), (9), (10), and (11) were substituted into (3) and the integration was performed, the model was rearranged into the following differential equation for calculating the degree of coalescence for the assumption of steady-state rheological properties:

$$8 \left(\lambda K_1 \frac{d\theta}{dt} \right)^2 + \left(-2\lambda K_1 + \frac{a_o \eta K_1^2}{\Gamma K_2} \right) \frac{d\theta}{dt} - 1 = 0, \quad (12)$$

where K_1 in their calculations is the small angle form of that defined in (10) and

$$K_2 = \frac{2^{-5/3} \cos \theta \sin \theta}{(1 + \cos \theta)^{4/3} (2 - \cos \theta)^{5/3}}. \quad (13)$$

Equation (12) predicts that the coalescence rate decreases with an increase in the relaxation time. The predicted behavior was used to suggest that viscoelasticity was responsible for the observation that two propylene ethylene copolymers coalesced at a slower rate than predicted by the Newtonian model (Bellehumeur et al. 1998). This is interesting because reported coalescence times for PTFE and several acrylic resins were significantly shorter than what was predicted by the Newtonian model (Mazur et al. 1995; Mazur and Plazek 1994). Furthermore, the relaxation times were not obtained from rheological data. Instead, they were adjusted to produce better agreement with data, and these values were unrealistically large. Therefore, it was unclear if it was possible to accurately predict coalescence using a viscoelastic constitutive equation with experimentally measured relaxation times.

Transient UCM model

To evaluate the importance of accurately representing the transient behavior, the UCM model was reformulated without the steady state assumption. The generalized UCM constitutive model is given by,

$$\tau_k + \lambda_k \left[\frac{\partial}{\partial t} \tau_k + \mathbf{v} \cdot \nabla \tau_k - \nabla \mathbf{v}^T \cdot \tau_k - \tau_k \cdot \nabla \mathbf{v} \right] = 2\eta_k \mathbf{D}, \quad (14)$$

where the subscript k signifies the mode or the k th partial stress component of the total stress, τ , and the total stress is the sum of the partial stresses, $\tau = \sum \tau_k$. Assuming that there is a homogeneous flow field and only one mode, (14) is reduced to,

$$\tau + \lambda \left[\frac{d}{dt} \tau - \nabla \mathbf{v}^T \cdot \tau - \tau \cdot \nabla \mathbf{v} \right] = 2\eta \mathbf{D}, \quad (15)$$

which differs from (11) because the transient term, $d\tau/dt$, is included. This change increases the complexity of the problem because the components of the extra stress tensor are not only differential equations, as shown in (16) and (17), but each contains the extension rate, which is also defined as a differential equation in (9):

$$\frac{d\tau_{xx}}{dt} = \frac{2\eta \dot{\epsilon}}{\lambda} + \tau_{xx} \left(2\dot{\epsilon} - \frac{1}{\lambda} \right), \quad (16)$$

$$\frac{d\tau_{yy}}{dt} = \frac{-4\eta \dot{\epsilon}}{\lambda} - \tau_{yy} \left(4\dot{\epsilon} + \frac{1}{\lambda} \right), \quad (17)$$

$$\tau_{zz} = \tau_{xx}. \quad (18)$$

As with the steady-state UCM model, these stress equations are coupled with (5), (6), (7), (9), (10), and

substituted into the energy balance, (3). It should be noted that the energy balance expressed by (3) is only valid for viscoelastic fluids with the assumption that the internal energy is independent of the configuration tensor and for this case involving surface tension the internal energy is only a function of surface area. This assumption is commonly made when modeling non-isothermal viscoelastic flow (Leonov 1987).

Numerical method and initial conditions

The system of equations describing coalescence is solved by substituting (4), (5), (6), (7), (9), and (10) into the energy balance, Eq. (3). This gives for all rheological models an equation in time of the form:

$$\frac{a_0 2^{2/3} K_1}{3\Gamma} (\tau_{xx} - \tau_{yy}) \frac{(1 + \cos \theta)^{4/3} (2 - \cos \theta)^{5/3}}{\cos \theta \sin \theta} - 1 = 0. \quad (19)$$

For the Newtonian and steady-state UCM model, the stresses, τ_{xx} and τ_{yy} , are known explicitly as functions of $\dot{\epsilon}$. For the transient single-mode UCM model, two additional ordinary differential equations need to be solved for the stresses. These are determined from (16) and (17) by assuming that the extension rate is constant over a given time step. This is a good approximation if the time step chosen is sufficiently small. Then the differential equations for the stresses, (16) and (17), may be solved analytically for $t_n \leq t \leq t_{n+1}$ to give,

$$\tau_{xx} = e^{-(\frac{1}{\lambda} - 2\dot{\epsilon})t} \left(\frac{2\eta\dot{\epsilon}}{(1 - 2\lambda\dot{\epsilon})} e^{(\frac{1}{\lambda} - 2\dot{\epsilon})t} + C_{xx} \right), \quad (20)$$

$$\tau_{yy} = e^{-(\frac{1}{\lambda} + 4\dot{\epsilon})t} \left(\frac{-4\eta\dot{\epsilon}}{(1 + 4\lambda\dot{\epsilon})} e^{(\frac{1}{\lambda} + 4\dot{\epsilon})t} + C_{yy} \right), \quad (21)$$

where the integration constants C_{xx} and C_{yy} are determined from the expressions:

$$C_{xx} = \left[\tau_{xx}(t_n) - \frac{2\eta\dot{\epsilon}}{(1 - 2\lambda\dot{\epsilon})} \right] e^{(\frac{1}{\lambda} - 2\dot{\epsilon})t_n}, \quad (22)$$

$$C_{yy} = \left[\tau_{yy}(t_n) + \frac{4\eta\dot{\epsilon}}{(1 + 4\lambda\dot{\epsilon})} \right] e^{(\frac{1}{\lambda} + 4\dot{\epsilon})t_n}. \quad (23)$$

At the beginning of a time step the value of the coalescence angle $\theta_n = \theta(t_n)$, is known, either from the initial condition or the previous time step. The corresponding rate of the coalescence angle at time t_n , $(d\theta/dt)_n$, is obtained by determining the root of (19) using Müller's method (IMSL 1997; Müller 1956). At the end of a time step, the stresses and extension rate are updated and used to determine the constants C_{xx} and C_{yy} for the next step. Finally, the new coalescence angle, θ_{n+1} , is determined using a forward difference formula.

The initial condition of x/a used to solve (8; Newtonian), (12; UCM steady state), or (19; UCM unsteady) was taken as 0.2, which was the value observed experimentally when the sample reached 180°C. The choice of the initial condition may be a major issue in accurately predicting coalescence of two polymeric drops. Pokluda et al. (1997) used the asymptotic Newtonian solution at $t = 0.0001\eta a_0/\Gamma$, which would have given a value of x/a of about 0.01 in this case. However, it was not experimentally possible to start with small values of the dimensionless neck radius. Part of this was due to some growth, which occurred at temperatures between when the drops reached the melting point ($\sim 165^\circ\text{C}$) and the test temperature of 180°C. It is also possible that the initial contact radius was due to elastic adhesive contact. The theory for the initial contact radius is based on balancing the work of adhesion against the work of elastic deformation and is referred to as the 'JKR' theory (Johnson et al. 1971). This theory was later extended to viscoelastic materials by Schapery (1989) and later by Hui et al. (1998) to give the following expression for the dimensionless contact radius:

$$\frac{x}{a} = \left(\frac{9\pi C_0 \Gamma}{a} \right)^{1/3}, \quad (24)$$

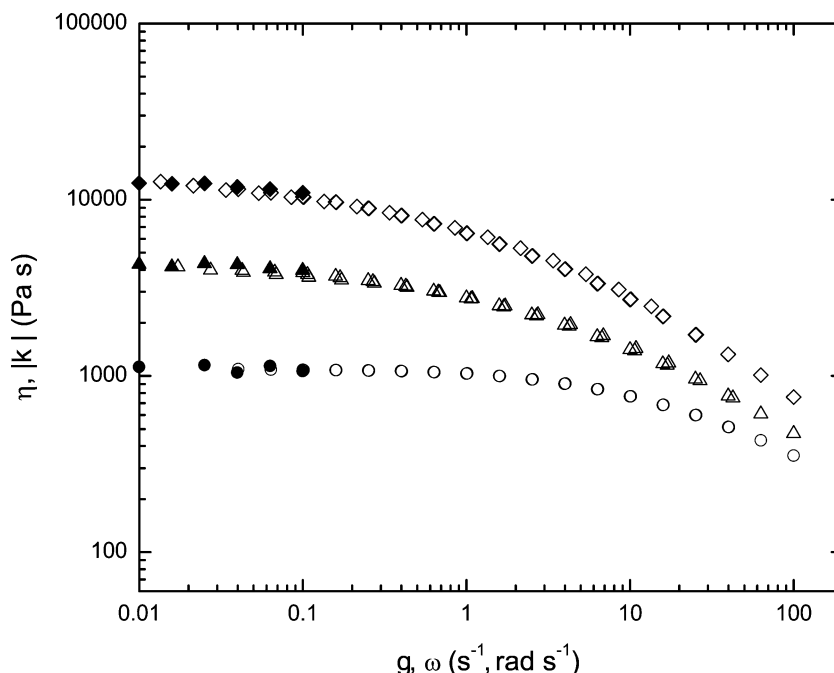
where C_0 is the equilibrium compliance (usually identified as J_e^0). However, based on measured values of J_e^0 for the three PPs (see Table 2) the initial dimensionless contact radii are predicted to be 1.29, 1.21, and 1.55, for the PP samples with increasing M_w , which are unreasonably high (as the maximum value is 1.0). Hence, initial values of x/a based on elastic contact cannot be used as the fluids are too compliant. At 165°C, J_e^0 could be low enough to promote some elastic contact, but this is a difficult number to obtain at this temperature because of crystallinity. This matter of initial conditions can be taken up for future study, but for the time being the emphasis is on isothermal viscous coalescence and the experimentally observed value will be used throughout the calculations.

Results and discussion

Newtonian and steady-state UCM models

Newtonian and UCM constitutive model parameters were obtained from the rheological data. The shear viscosity master curves for the three polypropylenes at 180°C are shown in Fig. 3. The steady and dynamic rheological data were used to identify an acceptable rate to measure the zero shear viscosity and the transient viscosity. As shown in the figure, each of the samples exhibit zero shear viscosity behavior at a shear rate of

Fig. 3 Steady and complex shear viscosity master curves for polypropylene at 180°C: (open circles) 190 k, (open triangles) 250 k, (open diamonds) 340 k. The open symbols represent small amplitude oscillatory shear measurements; filled symbols represent steady shear values



0.01 s^{-1} , and, therefore, that shear rate was used to obtain parameters for the single-mode UCM constitutive model. In addition, the value of the zero shear viscosity was used for the viscosity in the Newtonian model.

The single-mode UCM model fits to data at a shear rate of 0.01 s^{-1} as shown in Fig. 4. The model fits the steady-state viscosity well but some error does exist in the initial transient region. A summary of the coalescence model parameters and calculated Deborah numbers, as defined for coalescence, is provided in Table 2.

The results from the coalescence experiments are shown in Fig. 5 along with predictions from the Newtonian and steady-state UCM models using the parameters that were obtained experimentally. The coalescence data are in qualitative agreement with the suggestions of the Frenkel model; coalescence rates decrease with increasing viscosity. The Newtonian model predicts slower coalescence rates than what were experimentally measured. This result is consistent with the previously mentioned reports on PTFE and acrylic resins in which it was concluded that the experimental coalescence was faster than predicted by the Newtonian model (Mazur et al. 1995; Mazur and Plazek 1994).

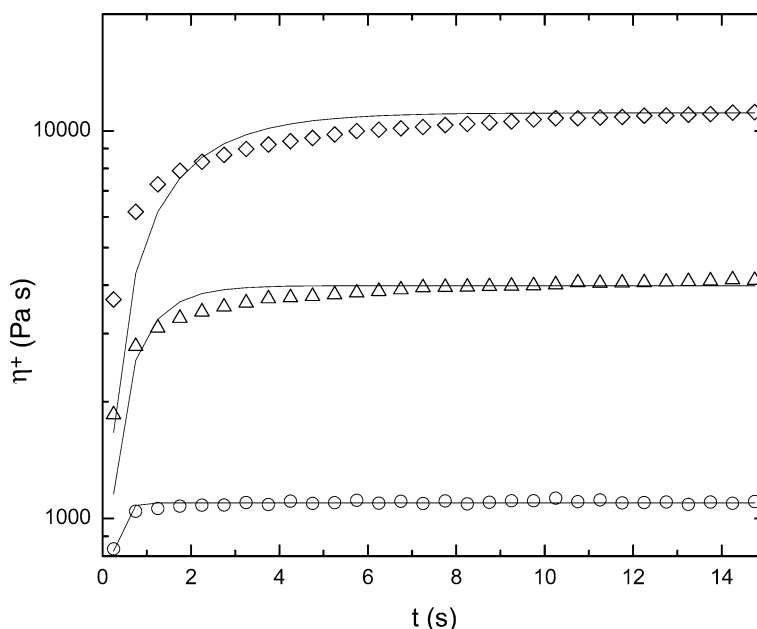
The inaccuracy of the Newtonian model at times longer than the characteristic relaxation time, especially for the 190 k and 250 k samples, has discouraging implications on the accuracy of viscoelastic model predictions at long times. Error between the experimental data and the predicted Newtonian behavior at long times will also be present in the UCM coalescence models because the UCM constitutive model predicts

Newtonian behavior at long times (this assumes that $\dot{\epsilon} \ll \dot{\epsilon}_c$ where $\dot{\epsilon}_c$ is the critical extension rate at which the UCM model predicts that the biaxial extensional viscosity approaches infinity). Interestingly, the steady-state model predictions are essentially identical to the Newtonian model predictions, overlapping not only at long times, but also at short times as shown in Fig. 5.

Following the approach by Bellehumeur et al. (1998), the relaxation time was arbitrarily increased in an attempt to improve the accuracy of the predictions. As illustrated in Fig. 6, the coalescence rates decrease at short times with large changes in the relaxation time. However, this only decreases the accuracy of the predictions by slowing coalescence rates relative to the predicted Newtonian limit. The facts that the steady-state UCM model predictions are the same as the Newtonian predictions for the values of relaxation time obtained from the experimental data, the magnitude of the relaxation times need to be relatively large to observe a difference from the Newtonian model, and increasing the relaxation time decreases accuracy, suggesting that the behavior predicted by the steady-state model may not be representative of the response of viscoelastic materials.

The reason that the coalescence rate of the steady-state model is predicted to be slow as the relaxation time is increased can be determined by considering the magnitude of the biaxial extension rates and the implications it has on the nature of the biaxial viscosity. The evolution of the biaxial extension rates during coalescence, as predicted by the steady-state UCM model for the three evaluated relaxation times, are shown in Fig. 7. The

Fig. 4 Single-mode UCM model fit to the transient shear viscosity from stress growth experiments at 180°C and $g = 0.01 \text{ s}^{-1}$. The *symbols* represent the experimental data: (*open circles*) 190 k, (*open triangles*) 250 k, (*open diamonds*) 340 k. The *lines* represent the single mode UCM fits to the data



extension rates pass through a maximum and then decay monotonically. The UCM model predicts that the biaxial extensional viscosity, $\eta_b(\dot{\epsilon}(t))$, approaches infinity at a critical extension rate, $\dot{\epsilon}_c = 1/2\lambda$. It is reasonable to suspect that the enhanced viscosity, which simulates a rate hardening behavior, slows coalescence excessively. The magnitude of the extension rates for the large relaxation times of 200 and 400 s do not exceed the

predicted critical extension rates of 0.0025 and 0.00125 s^{-1} , respectively, but they are close enough to cause the biaxial viscosity to increase beyond $6\eta_o$, as shown in Fig. 8. The biaxial extensional viscosity predicted for the experimentally measured relaxation time ($\lambda = 1.54 \text{ s}$) is practically identical to $6\eta_o$ and explains the similarity in coalescence rates for the steady-state UCM and Newtonian cases. By increasing the relaxation

Fig. 5 Experimental polypropylene coalescence data, the Newtonian, and steady-state UCM model predictions at 180°C. The *symbols* represent the experimental data: (*open circles*) 190 k, (*open triangles*) 250 k, (*open diamonds*) 340 k. The *lines* represent the model predictions: (*solid lines*) Newtonian model, (*dashed lines*) steady-state UCM model. Note that the predictions from the steady-state model coincide with the Newtonian model predictions

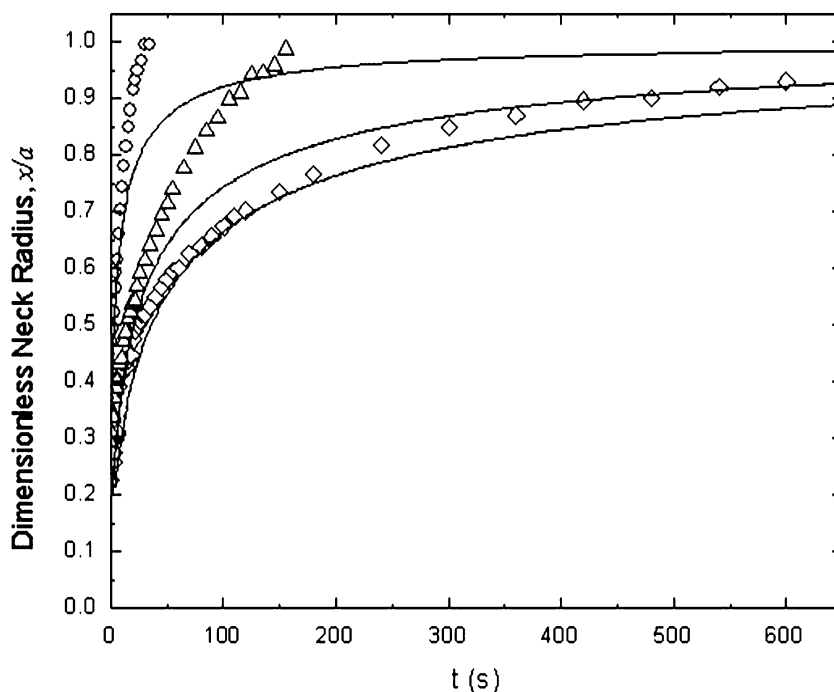
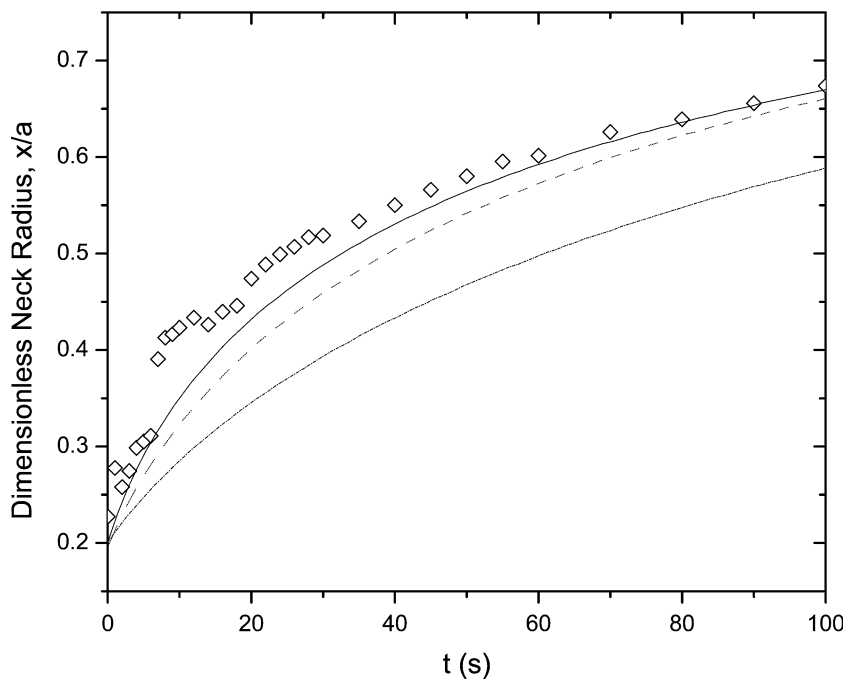


Fig. 6 340 k polypropylene coalescence data, the Newtonian, and steady-state UCM model predictions at short times (< 100 s). The *symbols* represent the experimental data (*open diamonds*). The *lines* represent the model predictions: (*solid lines*) Newtonian, steady-state UCM model with (*dashed lines*) $\lambda = 1.54$ s, (*dotted lines*) $\lambda = 200$ s, (*dashed dotted lines*) $\lambda = 400$ s



time, the biaxial extensional viscosity significantly surpasses $6\eta_0$ leading to a decrease in the coalescence rate relative to the Newtonian model. This explains why the steady-state model predicts a decrease in the coalescence rate as the relaxation time is increased.

Single-mode transient UCM model

Predictions were generated with the single-mode transient model to ascertain the role of the transient representation of material functions in accurately predicting

Fig. 7 Biaxial extension rate as predicted by the steady-state UCM model during coalescence of the 340 k sample where: (*solid lines*) $\lambda = 1.54$ s, (*dashed lines*) $\lambda = 200$ s, (*dotted lines*) $\lambda = 400$ s

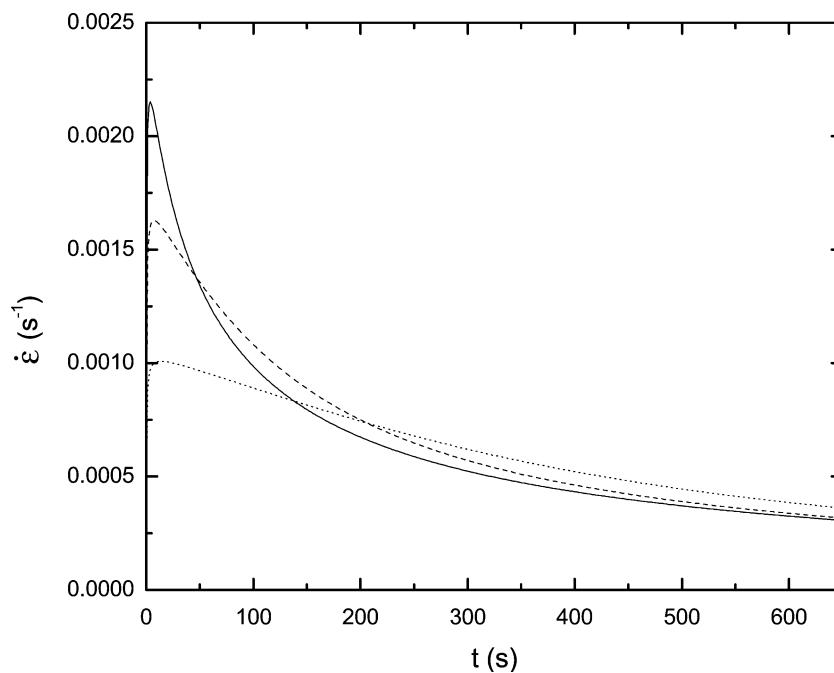
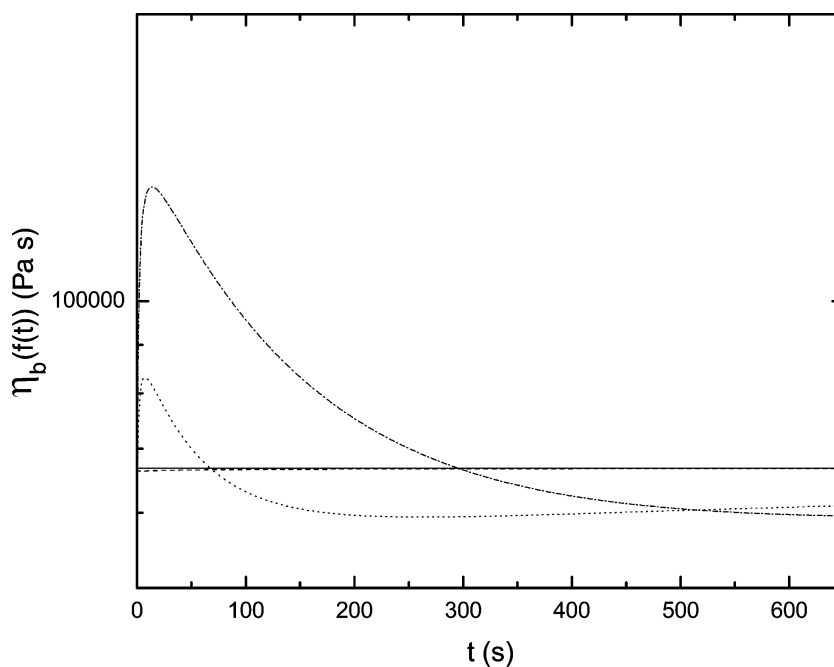


Fig. 8 Biaxial extensional viscosity as predicted by the steady-state UCM model during coalescence of the 340 k sample where: (*solid lines*) represents $6\eta_0$ that is predicted for Newtonian fluids, and (*dashed lines*) for $\lambda = 1.54$ s, (*dotted lines*) $\lambda = 200$ s, (*dashed dotted lines*) $\lambda = 400$ s



the coalescence rates of polymeric materials. Transient model predictions for the 340 k sample are compared with data in Fig. 9 using the relaxation time determined from experimental data and also, as before, two larger values. In addition, the Newtonian model prediction is

included for reference. The transient model predictions do not decrease the error that was observed at long times because the UCM model predicts Newtonian behavior at long times as previously discussed. However, it does appear that the transient model improves the accuracy

Fig. 9 340 k polypropylene coalescence data, the Newtonian, and transient UCM model predictions. The *symbols* represent the experimental data (*open diamonds*). The *lines* represent the model predictions: (*solid lines*) Newtonian, transient UCM model with (*dashed lines*) $\lambda = 1.54$ s, (*dotted lines*) $\lambda = 2.0$ s, (*dashed dotted lines*) $\lambda = 3.0$ s

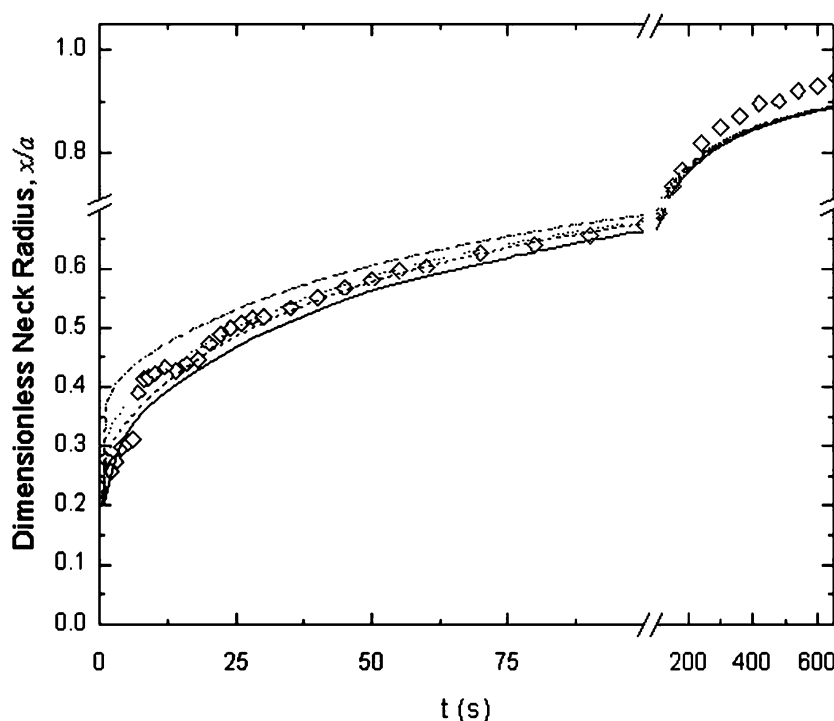
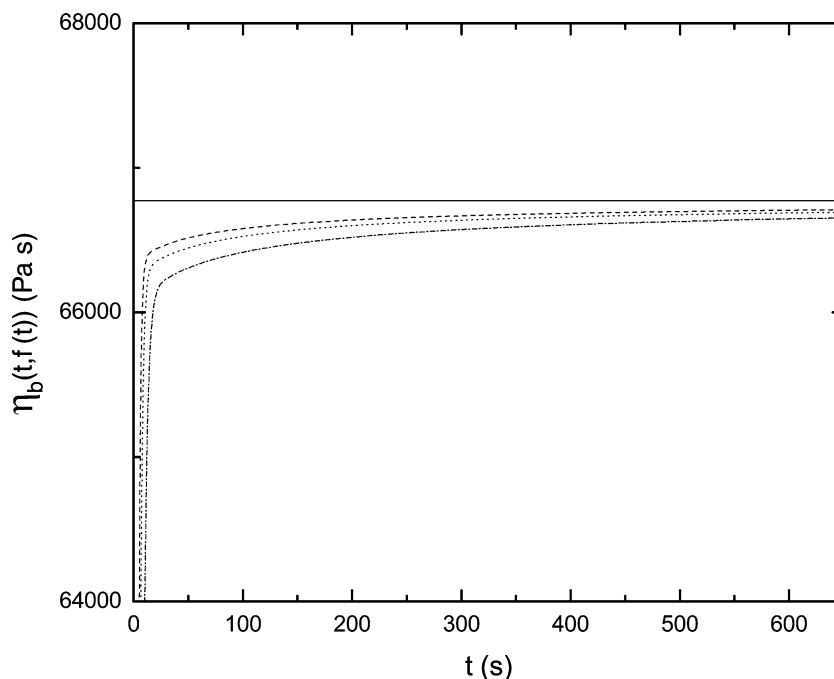


Fig. 10 Biaxial extensional viscosity as predicted by the transient UCM model during coalescence of the 340 k sample where: (*solid lines*) represents $6\eta_o$ that is predicted for Newtonian fluids, and (*dashed lines*) for $\lambda = 1.54$ s, (*dotted lines*) $\lambda = 200$ s, (*dashed dotted lines*) $\lambda = 400$ s



at shorter times (which can be important in processes, such as spray coating, where coalescence times are limited to shorter times).

Upon focusing on the shorter times where viscoelastic influence is relevant, it becomes apparent that the transient UCM model predicts the reverse effect of

relaxation time on coalescence when compared to the steady-state model. By using values for relaxation time that are greater than the experimentally measured value, the coalescence rate increases relative to the Newtonian model. This is in qualitative agreement with what has been reported for the PTFE and acrylic studies, where it

Fig. 11 Multimode fit to the storage and loss moduli for the 340 k sample at 180°C. (*open circles*) G' , (*open triangles*) G'' . The *symbols* represent the experimental data; the *lines* represent the multimode UCM fit to the data

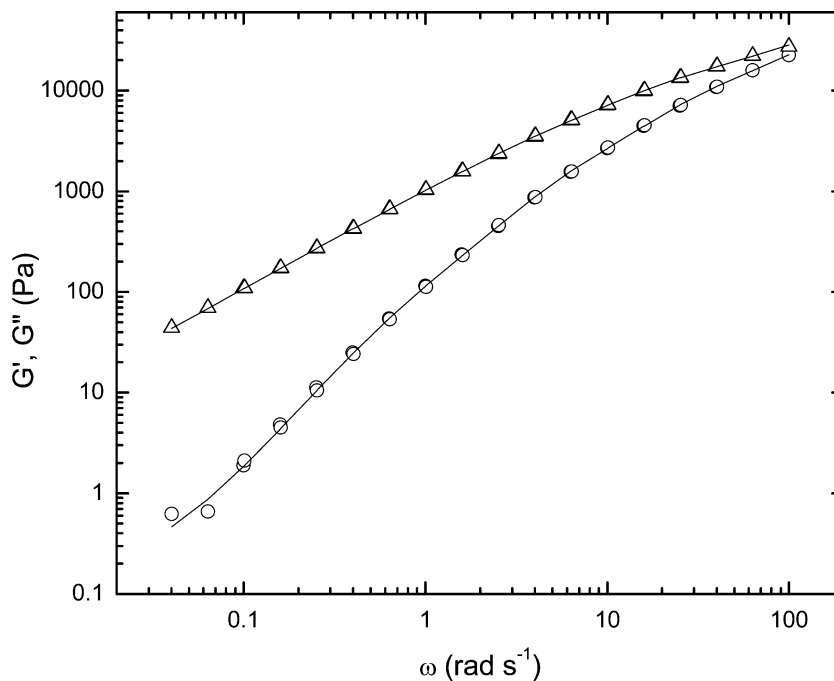
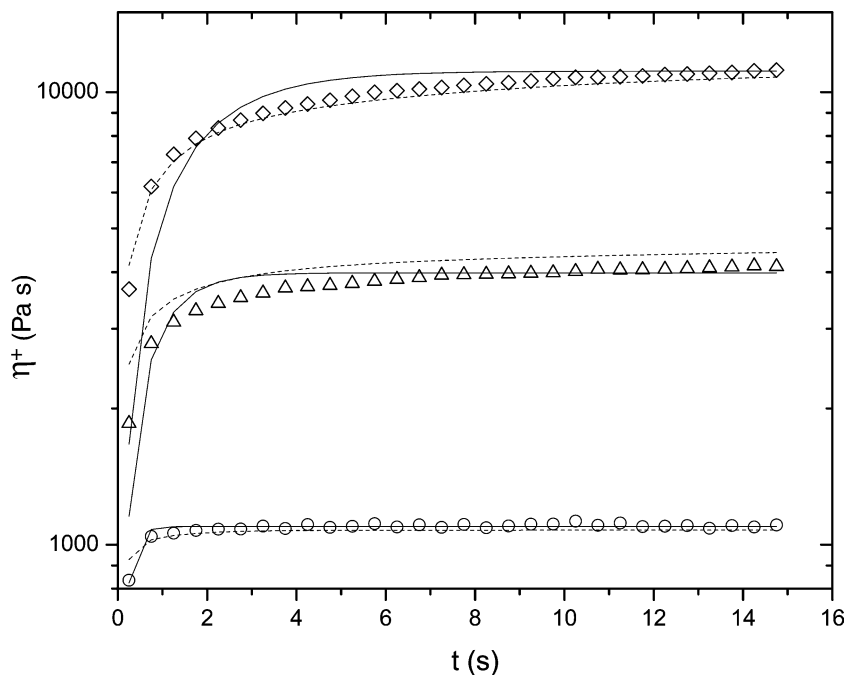


Fig. 12 Transient shear viscosity at 180°C and $g = 0.01 \text{ s}^{-1}$, the single-mode UCM model fits, and the multimode UCM model predictions. The *symbols* represent the experimental data: (*open circles*) 190 k, (*open triangles*) 250 k, (*open diamonds*) 340 k. The *lines* represent the models: (*solid lines*) single-mode fits, (*dotted lines*) multimode predictions



was reported that coalescence rates increase with increasing relaxation time (Mazur et al. 1995; Mazur and Plazek 1994). The transient model improves the accuracy of the coalescence predictions for these materials, producing quantitative agreement within

experimental error at short times. The transient model also appears to be more sensitive than the steady-state model to variations in the relaxation time and does not require unrealistically large values to affect the coalescence rate. Furthermore, the biaxial extensional viscosity

Fig. 13 340 k polypropylene coalescence data, the Newtonian, the single-mode transient UCM model, and the multimode transient UCM model predictions at short times. The *symbols* represent the experimental data (*open diamonds*). The *lines* represent the model predictions: (*solid lines*) Newtonian, (*dashed lines*) single-mode transient UCM model, (*dotted lines*) multimode transient model

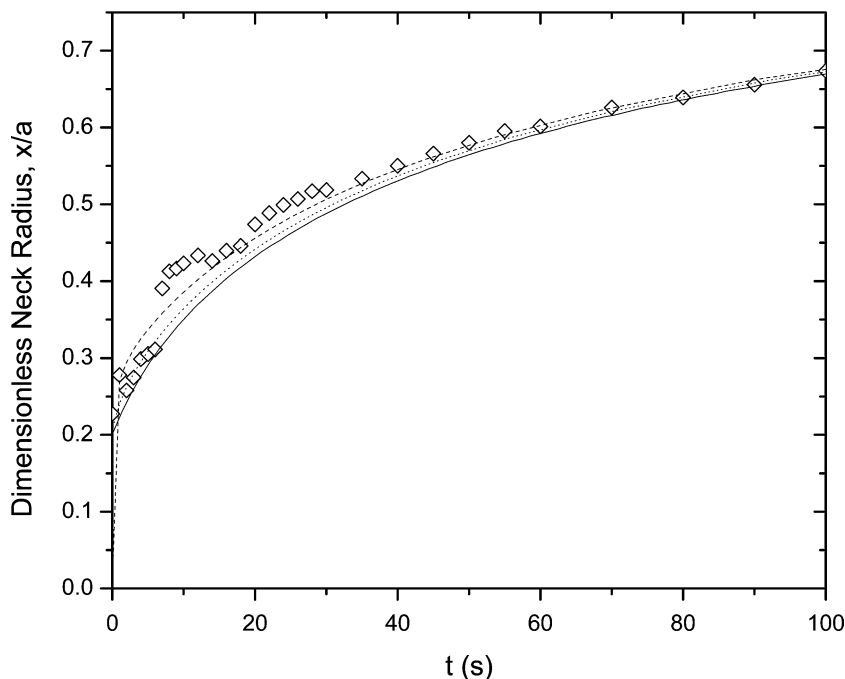


Table 3 Multimode UCM model parameters at 180°C

	k	1	2	3	4	5	6
190 k	η_k (Pa s)	287.9	381.4	310.5	97.03	91.05	—
	λ_k (s)	0.0040	0.0320	0.1722	1.033	465.2	—
250 k	η_k (Pa s)	800.2	997.7	1057	1084	607.3	839.4
	λ_k (s)	0.0024	0.0634	0.3355	1.600	10.16	761.3
340 k	η_k (Pa s)	1085	2678	3778	3260	1275	—
	λ_k (s)	0.0224	0.1629	1.011	6.485	49.20	—

approaches but does not exceed $6\eta_o$, as is shown in Fig. 10, thus eliminating the rate-hardening behavior predicted by the steady-state model.

Multimode transient UCM model

The previous results, shown in Fig. 9, were based on calculations using a single relaxation time. The multimode UCM model was used to further examine the importance of accurately representing the transient viscosity during coalescence. A representative example of the multimode UCM model fit to the storage and loss moduli is shown for the 340 k sample in Fig. 11. The multimode UCM model parameters are summarized in Table 3.

To establish the significance of using a relaxation spectrum, the model parameters were used to generate predictions for the stress growth data. These predictions are compared to the single-mode fits and experimental data in Fig. 12. The multimode model appears to be more accurate than the single-mode UCM model fits to the stress growth data, especially at short times, and for the highest molecular weight sample. In general, it is expected that the magnitude of the difference in predicted stress growth behavior between the single mode and multimode UCM model will increase in materials possessing greater relaxation times or broader relaxation spectra and, therefore, will become more important in the coalescence predictions for those cases.

The coalescence prediction for the 340 k sample, which is representative of what was observed for all samples, is shown along with the Newtonian and single mode transient UCM results in Fig. 13. At short times (less than 50 s), the multimode model exhibits a slight decrease in the coalescence rate relative to the single-mode model at short times. This reduction in coalescence rate seems reasonable when we consider the single-mode fit and multimode prediction of the transient viscosity that was shown in Fig. 12. The multimode model predicts a higher viscosity than the single mode at less

than approximately 2 s. This should slow coalescence relative to the single mode because the transient shear viscosity behaves similarly to the biaxial extensional viscosity. It is interesting that this seemingly minor difference at short times has such a dramatic effect on the magnitude of the coalescence rate over much longer times. This emphasizes that slight differences in the representation of the transient viscosity has a great influence on the early stages of coalescence and that the impact of the multimode model will be more significant with increasing viscoelasticity (higher Deborah numbers).

Conclusions

The coalescence model developed by Frenkel and Eshelby for Newtonian fluids was extended to viscoelastic fluids without the steady state approximation for the stresses using the transient form of the UCM with single- and multimode relaxation times. The single-mode transient coalescence model demonstrated that the influence of viscoelasticity is limited to short times and at long times it converges to the Newtonian solution. This formulation was able to improve the model accuracy at short times by predicting an increase in coalescence rate with an increase in relaxation time and in doing so illustrated the importance of representing the transient viscosity. The model was also more sensitive to the magnitude of the relaxation time, producing significant changes in coalescence rates with only small changes in the magnitude of the relaxation time. While the multimode transient UCM model more accurately represented the transient rheological response it was unable improve the accuracy of the model predictions for these materials, but will likely be more important for fluids possessing a broader relaxation spectrum.

Acknowledgements This work was financially supported by a phase II SBIR grant from NASA, grant number NAS-2S-4018-285, managed by Luna Innovations.

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